6.20 Solution: (a) In the Coulomb gauge,

$$\begin{aligned}
\Phi(\mathbf{x},t) &= \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{x}',t)}{|\mathbf{x}-\mathbf{x}'|}d^3x' = \frac{1}{4\pi\epsilon_0}\int \frac{\delta(x')\delta(y')\delta'(z')\delta(t)}{|\mathbf{x}-\mathbf{x}'|}d^3x' \\
&= \frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty}\frac{\delta'(z')}{[x^2+y^2+(z-z')^2]^{1/2}}dz' \\
&= -\frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty}\delta(z')\frac{\partial}{\partial z'}[x^2+y^2+(z-z')^2]^{-1/2}dz' \\
&= -\frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty}\delta(z')\frac{z-z'}{[x^2+y^2+(z-z')^2]^{3/2}}dz' \\
&= -\frac{1}{4\pi\epsilon_0}\delta(t)\frac{z}{r^3}.
\end{aligned}$$

(b) The transverse current is defined as

$$\mathbf{J}_t(\mathbf{x},t) = \frac{1}{4\pi}\nabla\times\nabla\times\int\frac{\mathbf{J}(\mathbf{x}',t)}{|\mathbf{x}-\mathbf{x}'|}d^3x'.$$

Since

$$\int\frac{\mathbf{J}(\mathbf{x}',t)}{|\mathbf{x}-\mathbf{x}'|}d^3x' = -\delta'(t)\frac{\boldsymbol{\epsilon}_3}{r},$$

the transverse current is then

$$\mathbf{J}_t(\mathbf{x},t) = -\frac{1}{4\pi}\delta'(t)\nabla\times\nabla\times\left(\frac{\boldsymbol{\epsilon}_3}{r}\right) = -\frac{1}{4\pi}\delta'(t)\left[\nabla\left(\nabla\cdot\frac{\boldsymbol{\epsilon}_3}{r}\right)-\nabla^2\left(\frac{\boldsymbol{\epsilon}_3}{r}\right)\right].$$

The first term can be written as

$$\nabla\left(\nabla\cdot\frac{\boldsymbol{\epsilon}_3}{r}\right) = \nabla\left(\boldsymbol{\epsilon}_3\cdot\nabla\left(\frac{1}{r}\right)\right) = \nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right),$$

and the second term is

$$\nabla^2\left(\frac{\boldsymbol{\epsilon}_3}{r}\right) = \boldsymbol{\epsilon}_3\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta(\mathbf{x})\boldsymbol{\epsilon}_3.$$

Therefore, the transverse current becomes

$$\mathbf{J}_t(\mathbf{x},t) = -\frac{1}{4\pi}\delta'(t)\left[\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)+4\pi\delta(\mathbf{x})\boldsymbol{\epsilon}_3\right] = -\delta'(t)\left[\boldsymbol{\epsilon}_3\delta(\mathbf{x})+\frac{1}{4\pi}\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)\right].$$

Notice that

$$\frac{\partial}{\partial z}\left(\frac{1}{r}\right) = -\frac{z}{r^3},$$

then

$$\begin{aligned}
\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) &= -\nabla\left(\frac{z}{r^3}\right) = -\boldsymbol{\epsilon}_i\partial_i\left(\frac{z}{r^3}\right) \\
&= \frac{3x_iz}{r^5}\boldsymbol{\epsilon}_i - \frac{\boldsymbol{\epsilon}_3}{r^3} = \frac{3\mathbf{x}(\boldsymbol{\epsilon}_3\cdot\mathbf{x})}{r^5} - \frac{\boldsymbol{\epsilon}_3}{r^3} \\
&= \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n})-\boldsymbol{\epsilon}_3}{r^3},
\end{aligned}$$

for $r \neq 0$.

To account for the singularity at $r = 0$, consider the following integral over a ball with radius R and center at the origin $\mathbf{x} = 0$,

$$\frac{1}{4\pi}\int \nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) d^3x = \frac{1}{4\pi}\oint \frac{\partial}{\partial z}\left(\frac{1}{r}\right)\mathbf{n}da = -\frac{1}{4\pi}\oint \frac{\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n})}{r^2}da = -\frac{1}{4\pi}\int \mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n})d\Omega = -\frac{1}{3}\boldsymbol{\epsilon}_3,$$

where we have used the result that

$$\int n_i n_j d\Omega = \frac{4\pi}{3}\delta_{ij}.$$

Together with the regular part, we have

$$\frac{1}{4\pi}\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) = -\frac{1}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) + \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n}) - \boldsymbol{\epsilon}_3}{4\pi r^3},$$

and finally,

$$\mathbf{J}_t(\mathbf{x},t) = -\delta'(t)\left[\frac{2}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) + \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n}) - \boldsymbol{\epsilon}_3}{4\pi r^3}\right].$$

(c) Using results from parts (a) and (b), we can verify Eq. (6.29). From part (a),

$$\frac{\partial\Phi}{\partial t} = -\frac{1}{4\pi\epsilon_0}\delta'(t)\frac{z}{r^3}.$$

Then,

$$\begin{aligned}
\frac{1}{c^2}\nabla\left(\frac{\partial\Phi}{\partial t}\right) &= -\frac{\mu_0}{4\pi}\delta'(t)\nabla\left(\frac{z}{r^3}\right) = \frac{\mu_0}{4\pi}\delta'(t)\nabla\left[\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right] \\
&= \mu_0\delta'(t)\left[-\frac{1}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) + \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n}) - \boldsymbol{\epsilon}_3}{4\pi r^3}\right] \\
&= -\mu_0\delta'(t)\left[\frac{1}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) - \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n}) - \boldsymbol{\epsilon}_3}{4\pi r^3}\right].
\end{aligned}$$

On the other hand,

$$\mathbf{J}_l = \mathbf{J} - \mathbf{J}_t = -\delta'(t)\left[\frac{1}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) - \frac{3\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n}) - \boldsymbol{\epsilon}_3}{4\pi r^3}\right],$$

which establishes Eq. (6.29). Therefore, we can use the transverse current alone to calculate the vector potential and the electromagnetic fields.

To this end, we need to solve Eq. (6.30),

$$\nabla^2\mathbf{A} - \frac{1}{c^2}\frac{\partial^2\mathbf{A}}{\partial t^2} = -\mu_0\mathbf{J}_t.$$

I found that the hints given by Jackson are not very useful, so I will use another method to perform the solution process. Introduce the four-dimensional Fourier transform for the vector potential and the transverse current,

$$\mathbf{A}(\mathbf{x},t) = \int\frac{d^3k}{(2\pi)^3}\int\frac{d\omega}{2\pi}\mathbf{A}(\mathbf{k},\omega)e^{i\mathbf{k}\cdot\mathbf{x}-i\omega t}, \quad \mathbf{J}_t(\mathbf{x},t) = \int\frac{d^3k}{(2\pi)^3}\int\frac{d\omega}{2\pi}\mathbf{J}_t(\mathbf{k},\omega)e^{i\mathbf{k}\cdot\mathbf{x}-i\omega t},$$

the equation becomes

$$\left(-k^2+\frac{\omega^2}{c^2}\right)\mathbf{A}(\mathbf{k},\omega)=-\mu_0\mathbf{J}_t(\mathbf{k},\omega),$$

and

$$\mathbf{A}(\mathbf{k},\omega)=\frac{\mu_0\mathbf{J}_t(\mathbf{k},\omega)}{k^2-\omega^2/c^2}.$$

Notice that

$$\int\delta'(t)e^{i\omega t}dt=\int e^{i\omega t}d\left(\delta(t)\right)=-i\omega\int e^{i\omega t}\delta(t)dt=-i\omega,$$

the Fourier transform for the transverse current is

$$\begin{aligned}\mathbf{J}_t(\mathbf{k},\omega) &= \int d^3x\int dt\mathbf{J}_t(\mathbf{x},t)e^{-i\mathbf{k}\cdot\mathbf{x}+i\omega t}\\ &= i\omega\int\left[\boldsymbol{\epsilon}_3\delta(\mathbf{x})+\frac{1}{4\pi}\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)\right]e^{-i\mathbf{k}\cdot\mathbf{x}}d^3x\\ &= i\omega\int\left[\boldsymbol{\epsilon}_3\delta(\mathbf{x})-\frac{k_z\mathbf{k}}{4\pi r}\right]e^{-i\mathbf{k}\cdot\mathbf{x}}d^3x\\ &= i\omega\left(\boldsymbol{\epsilon}_3-\frac{k_z\mathbf{k}}{k^2}\right),\end{aligned}$$

where we have performed integration by parts twice on the second term and have used the Fourier transform of the r^{-1} potential,

$$\int\frac{e^{-i\mathbf{k}\cdot\mathbf{x}}}{r}d^3x=\frac{4\pi}{k^2}.$$

Then,

$$\mathbf{A}(\mathbf{k},\omega)=-\mu_0c^2\frac{i\omega}{\omega^2-k^2c^2}\left(\boldsymbol{\epsilon}_3-\frac{k_z\mathbf{k}}{k^2}\right).$$

Now, we can perform the Fourier transform to obtain the vector potential in the real space,

$$\mathbf{A}(\mathbf{x},t)=-\mu_0c^2\int\frac{d^3k}{(2\pi)^3}e^{i\mathbf{k}\cdot\mathbf{x}}\left(\boldsymbol{\epsilon}_3-\frac{k_z\mathbf{k}}{k^2}\right)\int\frac{d\omega}{2\pi}\frac{i\omega}{\omega^2-k^2c^2}e^{-i\omega t}.$$

Similar to Section 12.11, using the same integration contour, the frequency integral is given by

$$\begin{aligned}\int\frac{d\omega}{2\pi}\frac{i\omega}{\omega^2-k^2c^2}e^{-i\omega t} &= \frac{1}{2\pi}(-2\pi i)\mathrm{Res}\left[\frac{i\omega}{\omega^2-k^2c^2}e^{-i\omega t}\right]\\ &= (-i)\left[\frac{ikc}{2kc}e^{-ikct}+\frac{-ikc}{-2kc}e^{ikct}\right]\\ &= \cos(kct).\end{aligned}$$

Then, the vector potential becomes

$$\begin{aligned}\mathbf{A}(\mathbf{x},t) &= -\frac{\mu_0c^2}{8\pi^3}\int\left(\boldsymbol{\epsilon}_3-\frac{k_z\mathbf{k}}{k^2}\right)e^{i\mathbf{k}\cdot\mathbf{x}}\cos(kct)d^3k\\ &= -\frac{\mu_0c^2}{8\pi^3}\boldsymbol{\epsilon}_3\int e^{i\mathbf{k}\cdot\mathbf{x}}\cos(kct)d^3k-\frac{\mu_0c^2}{8\pi^3}\left(\nabla\frac{\partial}{\partial z}\right)\int\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{k^2}\cos(kct)d^3k.\end{aligned}$$

The first integral can be calculated as

$$\begin{aligned}
\int e^{i\mathbf{k}\cdot\mathbf{x}}\cos(kct)d^3k &= 2\pi\int_0^{+\infty}k^2dk\int_{-1}^{1}d(\cos\theta)e^{ikr\cos\theta}\cos(kct) \\
&= 2\pi\int_0^{+\infty}k\cos(kct)\frac{e^{ikr}-e^{-ikr}}{ir}dk \\
&= \frac{2\pi}{r}\int_{-\infty}^{+\infty}(-ik)e^{ikr}\cos(kct)dk \\
&= \frac{\pi}{r}\int_{-\infty}^{+\infty}(-ik)\left(e^{ik(r+ct)}+e^{ik(r-ct)}\right)dk \\
&= -\frac{\pi}{r}\frac{\partial}{\partial r}\int_{-\infty}^{+\infty}\left(e^{ik(r+ct)}+e^{ik(r-ct)}\right)dk \\
&= -\frac{2\pi^2}{r}\left[\delta'(r+ct)+\delta'(r-ct)\right].
\end{aligned}$$

By the same argument as in Section 12.11, we can see only the $\delta'(r-ct)$ term will contribute. Thus, the contribution to the vector potential from the first integral is

$$-\frac{\mu_0c^2}{8\pi^3}\boldsymbol{\epsilon}_3\int e^{i\mathbf{k}\cdot\mathbf{x}}\cos(kct)d^3k = \frac{\mu_0c^2}{4\pi r}\boldsymbol{\epsilon}_3\delta'(r-ct).$$

Similarly, the second integral is

$$\begin{aligned}
\int\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{k^2}\cos(kct)d^3k &= \frac{2\pi}{r}\int_{-\infty}^{+\infty}\frac{e^{ikr}}{ik}\cos(kct)dk \\
&= \frac{\pi}{r}\int_{-\infty}^{+\infty}\frac{1}{ik}\left(e^{ik(r+ct)}+e^{ik(r-ct)}\right)dk.
\end{aligned}$$

Again, we only keep the term that depends on $r-ct$, and notice that the inverse Fourier transform of $(i\omega)^{-1}$ is related to the Heviside function $\Theta(t)$, *i.e.*,

$$\int_{-\infty}^{+\infty}\left(-\frac{1}{i\omega}\right)e^{-i\omega t}\frac{d\omega}{2\pi} = \Theta(t).$$

Then,

$$\int\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{k^2}\cos(kct)d^3k = -\frac{2\pi^2}{r}\Theta(ct-r).$$

Putting everything together, the vector potential becomes

$$\begin{aligned}
\mathbf{A}(\mathbf{x},t) &= \frac{\mu_0c^2}{4\pi r}\boldsymbol{\epsilon}_3\delta'(r-ct)+\frac{\mu_0c^2}{4\pi}\left(\nabla\frac{\partial}{\partial z}\right)\left[\frac{\Theta(ct-r)}{r}\right] \\
&= \frac{1}{4\pi\epsilon_0 r}\boldsymbol{\epsilon}_3\delta'(r-ct)+\frac{1}{4\pi\epsilon_0}\left(\nabla\frac{\partial}{\partial z}\right)\left[\frac{\Theta(ct-r)}{r}\right].
\end{aligned}$$

The electric field can be found by the relation $\mathbf{E} = -\partial\mathbf{A}/\partial t$. Here, $\delta(x)$ and $\delta''(x)$ are even and $\delta'(x)$ is odd, we will have

$$\frac{\partial}{\partial t}\delta'(r-ct) = -\frac{\partial}{\partial t}\delta'(ct-r) = -c\delta''(ct-r) = -c\delta''(r-ct).$$

Also,

$$\frac{\partial}{\partial t}\Theta(ct-r) = c\delta(ct-r) = c\delta(r-ct),$$

then,

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\frac{c}{r}\boldsymbol{\epsilon}_3\delta''(r-ct) - \frac{c}{4\pi\epsilon_0}\left(\nabla\frac{\partial}{\partial z}\right)\left[\frac{\delta(r-ct)}{r}\right].$$

It is straightforward to verify that

$$\begin{aligned}\left(\nabla\frac{\partial}{\partial z}\right)\left[\frac{\delta(r-ct)}{r}\right] &= \nabla\left[\frac{z}{r^2}\delta'(r-ct) - \frac{z}{r^3}\delta(r-ct)\right] \\ &= \frac{z\mathbf{x}}{r^3}\delta''(r-ct) + \frac{\boldsymbol{\epsilon}_3}{r^2}\delta'(r-ct) - \frac{2z\mathbf{x}}{r^4}\delta'(r-ct) \\ &\qquad -\frac{z\mathbf{x}}{r^4}\delta'(r-ct) - \frac{\boldsymbol{\epsilon}_3}{r^3}\delta(r-ct) + \frac{3z\mathbf{x}}{r^5}\delta(r-ct) \\ &= \frac{z\mathbf{x}}{r^3}\delta''(r-ct) - \left(\frac{3z\mathbf{x}}{r^2} - \boldsymbol{\epsilon}_3\right)\left(\frac{\delta'(r-ct)}{r^2} - \frac{\delta(r-ct)}{r^3}\right).\end{aligned}$$

Now, the electric field becomes

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\frac{c}{r}\left[\left(\boldsymbol{\epsilon}_3 - \frac{z\mathbf{x}}{r^2}\right)\delta''(r-ct) + \left(\frac{3z\mathbf{x}}{r^2} - \boldsymbol{\epsilon}_3\right)\left(\frac{\delta'(r-ct)}{r} - \frac{\delta(r-ct)}{r^2}\right)\right],$$

which agrees with the component-wise form given in the problem. Clearly, the electric field is causal, even the scalar potential in the Coulomb gauge breaks causality.

In the scanned solution, I have also performed the calculation in Lorenz gauge, and no surprise, the potentials and the fields are causal, and the electric field is the same as calculated in the Coulomb gauge, as it should be.